Cross-displinary applications of the Toulbar2 solver

UE : Introductory Research Project

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Date : March 31, 2026

Abstract

resumé à mettre ici

Key Words : *Informatics, Toulbar2, OSAC, VAC, Protein design, AI*

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List of Acronyms

CSP Constraint Satisfaction Problem

WCSP Weighted Constraint Satisfaction Problem

VCSP Valued Constraint Satisfaction Problem

CFNs Cost Function Networks

EPTs Equivalence Preserving Transformations

SAC Soft Arc Consistency

FDAC Full Directional Arc Consistency

EDAC Existential Directional Arc Consistency

VAC Virtual Arc Consistency

OSAC Optimal Soft Arc Consistency

CP Constraint Programming

LP Linear Programming

ILP Integer Linear Programming

MILP Mixed-Integer Linear Programming

MINLP Mixed Integer Non-Linear Programming

MIP Mixed Integer Programming

KPCG knapsack problem with a conflict graph

SBF Smart Brute-force

ZP Zahn's problem

BnB Branch-And-Bound

Introduction

The Constraint Satisfaction Problem (CSP) involves finding an assignment for a set of variables over a finite domain that respects a set of given hard constraints. To find a solution to a CSP problem, backtracking search is used. This strategy will create a new subtree every time a value is assigned to a variable. Once the current instance is infeasible, the algorithm will backtrack and continue with a different assignment until a solution to the original problem is found. However, many real-world problems require optimization rather than mere satisfaction.

The Valued Constraint Satisfaction Problem (VCSP) or Weighted Constraint Satisfaction Problem (WCSP) is considered a generalization of the classic CSP. In a VCSP, a Cost Function Networks (CFNs) framework is used. CFNs are a tuplet consisting of a finite set of discrete variables X, a finite domain D, a finite set of soft constraint C, and a maximum cost k. Soft constraints assign a cost to a specific assignment of a subset of variables. The NP-Hard problem of the VCSP is to find the instantiation of minimal cost.

To solve a VCSP, the BnB algorithm is used. BnB explores the solution space by '*branching*' : it assigns a value to a variable and creates a new subtree. At any node, if the current instance has cost higher than maximum cost k or its cost is higher than the cost of the best solution found so far, the branch is pruned and the search backtracks. This process continues until all branches have been pruned, and the best global solution is found.

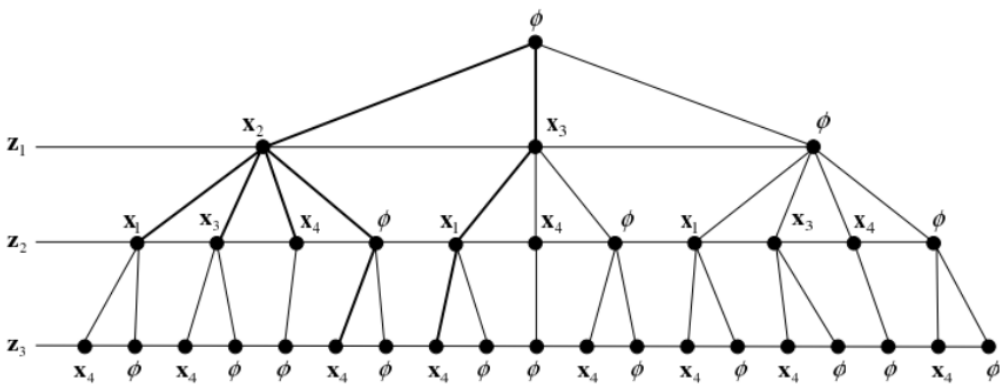

Figure 1: **Example of BnB : Mobile Robot Localisation and Mapping in Extensive Outdoor Environments.**[4]

The efficacy of the branch-and-bound depends on the lower bound we have access to. In a Cost Function Network, functions are categorized by their arity: unary functions (arity 1) assign costs to individual variable assignments, binary functions (arity 2) assign costs to pairs of variables, and the nullary constant (arity 0), denoted as $c_\emptyset$, represents a global lower bound on the cost of any complete assignment.

Equivalence Preserving Transformations (EPTs) transfer costs from one constraint to another while preserving the same optimal solution for the problem. Below, three different EPTss are presented for an example with three variables (x, y, z) with two possible assignments (a, b). A line joining two values represents a cost of 1 for this assignment in the corresponding binary cost function. Unary costs are represented as numbers inside the different assignments.

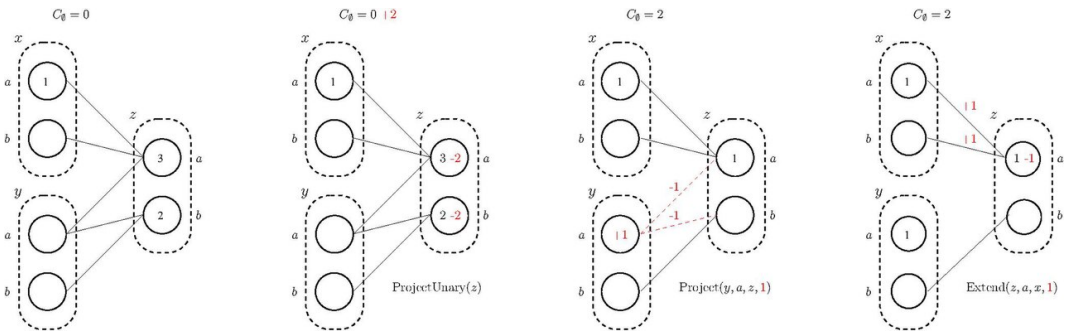

Figure 2: **Basic Equivalence Preserving Transformations.**[12]

By successive EPTs, the cost can be shifted and concentrated into the nullary constraint c, thus giving a better lower bound for the VCSP and allowing to quickly prune subtrees.

This process of applying successive EPTss to obtain a better lower bound is known as Soft Arc Consistency (SAC). Different inference techniques to obtain a good lower bound exist, and those include the following : Full Directional Arc Consistency (FDAC), Existential Directional Arc Consistency (EDAC), Virtual Arc Consistency (VAC) and Optimal Soft Arc Consistency (OSAC). Those inference techniques vary in relative strength vs relative computational cost, but generally the better the lower bound found, the higher the computational cost becomes. OSAC provides the best lower bound, followed by VAC, then EDAC and finally FDAC.

These theoretical findings are used in the solver Toulbar2. It is an open-source *C++* solver for discrete optimization problems within the cost function network framework. The solver integrates SAC operations, dynamically switching and choosing the best inference technique at a given node, and uses a branch-and-bound search to quickly explore the solution space and find the optimum. This document provides a state-of-the-art review of the solver's cross-disciplinary impact. We begin by examining the solver's ability to solve difficult combinatorial computer science problems against other state-of-the-art solvers, then explore their real world applications ranging from structural bioinformatics to classical operations research and finally discuss their recent integration into neuro-symbolic Artificial Intelligence networks.

1 Algorithmic Foundations & Solver Development

1.1 An application to graph theory and the cluster editing problem

Zahn's decision problem can be described as the cluster editing problem. Given a graph, what is the minimal number of operations (adding edges, removing edges) needed to create a disjoint union of cliques. The Zahn's problem (ZP) is generally an NP-hard problem. In a paper titled "A constraint programming approach to the Zahn's decision problem"[2], the ZP was transformed into a valued constraint satisfaction problem. This transformation is done as follows :

- A boolean variable is added for each pair of vertices. When a variable is set to true by the solver, then the decision is to keep or add the edge between the vertices. Else, the decision is to remove or not add an edge between the two vertices.
- A unary soft constraint is added for each variable. The cost of assigning 0 to a variable x is 1 if the edge exists in the original graph, and 0 otherwise.
- A ternary hard constraint is added for each pair of 3 edges which together creates the subgraph $P3$ described in the figure below. These ternary hard constraints are added to ensure that the resulting graph is a disjoint union of cliques. This result comes from the fact that a disjoint union of cliques cannot contain a subgraph isomorphic to $P3$, as explained in the paper.

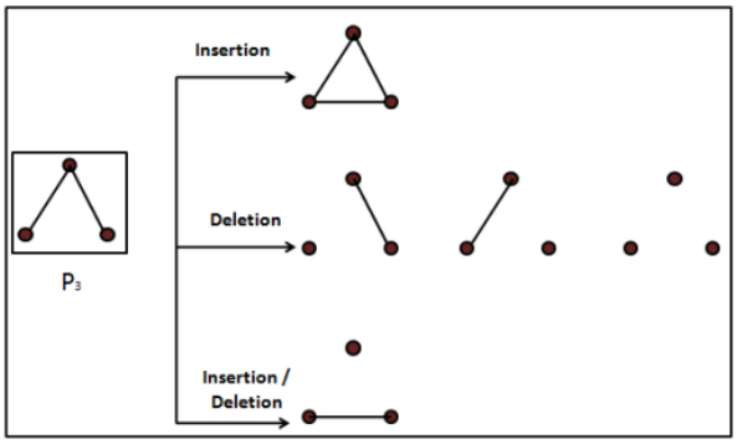

Figure 3: $P3$ **Graph and insertion/deletion possibilities.**[2]

Using Toulbar2 to solve the VCSP, the results found were much better than the fixed parameter algorithm for cluster editing presented before. The VCSP algorithm performed better on both sparse and dense graphs. These results show the efficacy and utility of the constraint programming paradigm for general graph related problems.

1.2 Constraint programming for calculating partition functions and discrete integration in high dimensions

Discrete integration in high dimensions is considered to be a generally unsolved and difficult problem, is typically #P-Hard and has applications to a wide variety of fields. This difficulty arises from the phenomenon known as the curse of dimensionality. In the paper "Taming the Curse of Dimensionality : Discrete Integration by Hashing and Optimization"[7], reduces the task of discrete integration into a collection of Maximum a Posteriori (MAP) sub-problems, which are NP-Hard instead.

The main challenge involves calculating the partition function. A partition function is a sum of all possible configurations of a set X where $w(x)$ is a weight function assigning a weight to a configuration.

$$Z = \sum_{x \in \mathcal{X}} w(x) \tag{1}$$

A commonly known example in physics is the Boltzmann distribution, a probability distribution that describes the likelihood of a system being in a specific configuration. The corresponding partition is as follows:

$$w(x) = e^{-\frac{E(x)}{k_B T}} \qquad \text{and} \qquad Z = \sum_{x \in \mathcal{X}} e^{-\frac{E(x)}{k_B T}} \tag{2}$$

- $E(x)$: The total energy of the configuration x.
- k_B: The Boltzmann constant.
- T: The absolute temperature.

A partition function can also be understood as a discrete integral, and is a computational task that is intractable due to the exponential number of configurations. WISH is a randomized algorithm which bypasses this challenging problem by solving, like said above, a collection of NP-Hard problems instead.

Using the state-of-the-art combinatorial tool Toulbar2 to solve the MAP queries, WISH provided a constant-factor approximation of discrete integrals with high probability, that could since then only be handled heuristically. Another significant advantage of this approach is that these independent optimization instances can be solved in parallel.

These results show the impact of Toulbar2 when a difficult #P problem is reduced to a combinatorial problem that can be solved with constraint programming. The wide applications of discrete integrations demonstrate the utility of the VCSP solver.

1.3 Toulbar2 for the weighted N-queen and quadratic assignment problems

The N-Queen problem is a non-trivial mathematical problem that looks for a placement of N queens on board of chess of size N by N where no two queens are threatening each other. The figure below shows an example of a solution where n = 8.

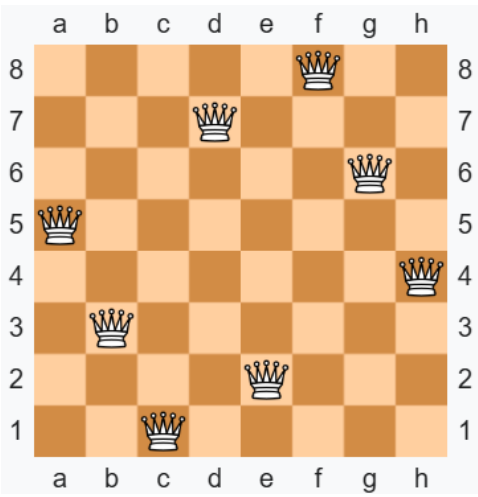

Figure 4: Example of a solution for the N-queen problem with n=8 [14]

The weighted N-Queen problem is a difficult optimization problem where every queen placement incurs a cost, and the goal is to find the lowest cost solution.

A key advancement by Sewa et al. (2022)[13] improved the Toulbar2 performance by integrating a more powerful way to propagate ALLDIFFERENT constraints for SAC. An ALLDIFFERENT constraint is a constraint where no two variables can have the same value. By modeling the cost as a unary constraint and using three ALLDIFFERENT constraints for the column, diagonal and anti-diagonal uniqueness, the upgraded version of Toulbar2 significantly outperformed other state-of-the-art solvers for the N-Queen problem.

Furthermore, with this new improved version of Toulbar2, the authors also tested its performance for the quadratic assignment problem (QAP). The QAP is a difficult NP-Hard problem where we seek to assign n facilities to n locations so as to minimize the total cost which is determined by the flow between facilities and the distance between locations. The results found that toulbar2 returned the highest quality solution with a gap of 3.9

These examples show the relevance of Toulbar2 when paired with a new propagator against state-of-the-art solvers for difficult NP-Hard combinatorial problems.

1.4 Toulbar2 application for the knapsack problem with a conflict graph and computational protein design

The knapsack problem is a classic NP-hard problem in combinatorics and computer science goes as follows : given a maximum weight capacity C, a set of objects each an assigned *weight* and *value*, what is the set of objects that maximize the value while respecting the weight capacity.

The knapsack problem with a conflict graph (KPCG) is a variant of the knapsack problem where conflict between objects is added. For a given graph G, each object is assigned to a vertex. An edge between vertex i and j implies that object i and j cannot be in a solution together. The KPCG is difficult to solve with constraint programming as CFNs do not allow the modeling of linear constraint without a non-trivial performance penalty, especially those with large coefficients.

Montalbano et al. (2022)[11] developed a way to model pseudo boolean linear constraints with CFNs allowing more problems to be solved using VCSP. When toulbar2 was compared with the state-of-the-art Integer Linear Programming (ILP) solver *CPLEX 20.1* for solving the KPCG. The approach was evaluated across six distinct classes of the Knapsack Problems with Conflict Graphs (KPCG), each having 720 instances with varying capacities, weight distributions, and profit correlations. The results found that Toulbar2 was more efficient than *CPLEX* in 4 out 6 classes. Furthermore, Toulbar2 found the best solutions for the largest number of instances in every class.

	TB2	CPLEX_t	CPLEX_d	TB2	CPLEX_t	CPLEX_d
C1	718	689	**720**	**720**	701	**720**
C3	597	487	**614**	**718**	513	639
C10	**490**	318	457	**633**	346	547
R1	**720**	705	**720**	**720**	705	**720**
R3	**720**	573	682	**720**	589	691
R10	**571**	365	519	**665**	384	583

Table 1: **Number of solved instances (left) and number of times a solver found the best solution within the time limit (right) for six different classes of KPCG.** Cplex was tested with two different encodings: tuples (t) and direct (d).[11]

This new technique also helped model the protein design problem, which is one of the main applications of the Toulbar2 solver, as it will be shown in the following section.

2 Bioinformatics & Life Sciences

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2.4 Conclusion

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3 Resource Management & Planning

Although the Toulbar2 solver has gained a strong reputation in the field of bioinformatics, the Cost Function Network formalism is proving particularly well-suited to classical operational research problems. The management and allocation of resources generally involve satisfying strict rules (hard constraints) whilst optimising quality criteria or preferences (soft constraints). This section illustrates how soft local consistency algorithms (such OSAC & EDAC) implemented in Toulbar2 enable the resolution of industrial-scale planning problems, whilst highlighting the strengths and limitations of these exact approaches compared to other paradigms such as Mixed-Integer Linear Programming (MILP).

3.1 Spatio-temporal allocation and conflict management

Planning and allocating physical resources often require merging two complex dimensions : space and time. This could mean rotating crops across agricultural land over several years. It could also involve coordinating flight paths in restricted airspace. These problems require satisfying strict rules while simultaneously optimizing quality criteria.
This subsection illustrates how the WCSP formalism models these spatio-temporal environments. It also evaluates the efficiency of the Toulbar2 solver compared to other exact methods.

The crop allocation problem consists of assigning different crop types (the values) to agricultural plots (the variables) over a finite time horizon. The approach proposed by Akplogan et al. (2013)[1] stands out by simultaneously modeling spatial and temporal constraints at the farm level.

The model integrates two types of constraints:

- **Hard constraints :** These guarantee agronomic feasibility. Examples include the minimum return time for a crop on the same plot, or respecting the biophysical properties of the soil.
- **Soft constraints :** These reflect the farmer's preferences. Examples include the spatio-temporal balance of the crops, or the beneficial effects of previous crops on the soil.

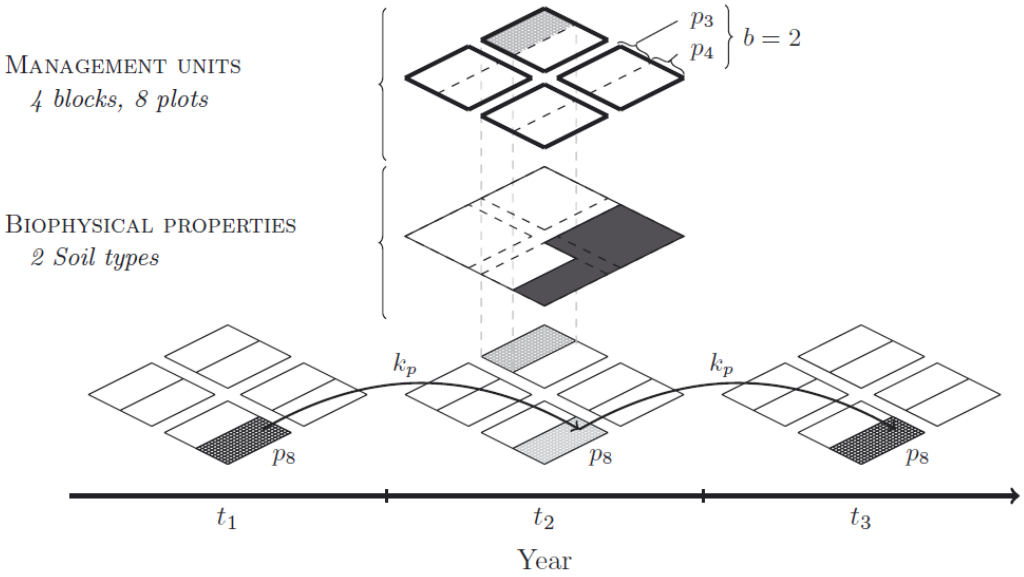


Figure 5: **Schematic representation of the spatial and temporal aspects of the decision–making problem** (t_i : year, b : block, p_j : plot, k_p : preceding effect).[1]

To solve this problem, the authors compared two approaches. The first is a WCSP formulation solved by Toulbar2. The second is a MILP reformulation solved by SCIP[1].
On a theoretical level, this study highlights a link with local consistency algorithms. The continuous relaxation of the MILP corresponds to the dual problem of the Linear Programming (LP) encoded by OSAC. When the upper bound is infinite, this relaxation is theoretically stronger than the EDAC consistency used by default in Toulbar2. However, if this upper bound decreases to a finite value, the two algorithms become incomparable.
However, the experiments reveal a major limitation of WCSP-based methods in this specific context. On real-world instances corresponding to medium and large farms, the SCIP solver proved nearly three times faster than Toulbar2. Furthermore, it consumed significantly fewer computational resources.

This study demonstrates a clear conclusion. WCSPs offer excellent flexibility for modeling complex spatio-temporal constraint networks. However, on large-scale, pure scheduling problems, the Toulbar2 solver faces real competition from classic MILP solvers.

In a much more dynamic spatio-temporal context, Schmidt et al. (2020)[5] apply the WCSP formalism to air traffic conflict resolution. A collision risk may be detected between several aircraft. When this happens, a generation method called "Lumberjack" produces a vast set of discretized alternative trajectories for each plane. The combinatorial challenge is then to select the best combination of compatible flight paths. This ensures the conflict is resolved optimally.

[1]For Mixed Integer Programming (MIP) and Mixed Integer Non-Linear Programming (MINLP), SCIP is now among the quickest non-commercial solvers. Additionally, it is a framework for branch-cut-and-price and constraint integer programming.

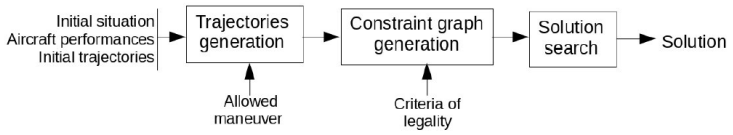


Figure 6: **Lumberjack method steps.**[5]

In this model, each aircraft represents a variable of the problem, and the set of its alternative avoidance trajectories constitutes its domain of values.

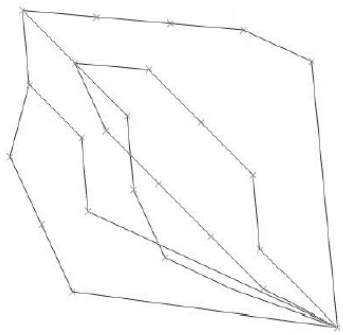


Figure 7: **Potential trajectories going from upper left to lower right.**[5]

The study compares the performance of the exact solver Toulbar2 with a custom-built "Smart Brute-force (SBF)" algorithm. Analyzing the results highlights a characteristic trait of the CFNs architecture and Toulbar2. The solver handles scaling the number of variables perfectly well. It efficiently resolves conflicts involving a relatively high number of aircraft (6 to 8 planes). In contrast, the SBF approach collapses under these conditions.
However, Toulbar2's memory saturates quickly when given too many possible trajectories per plane. In other words, it struggles if the domain size becomes too large. This issue can be explained by two factors. First is the memory cost of representing constraints in extension. Second is the heavy cost of applying local consistency algorithms to very large domains.

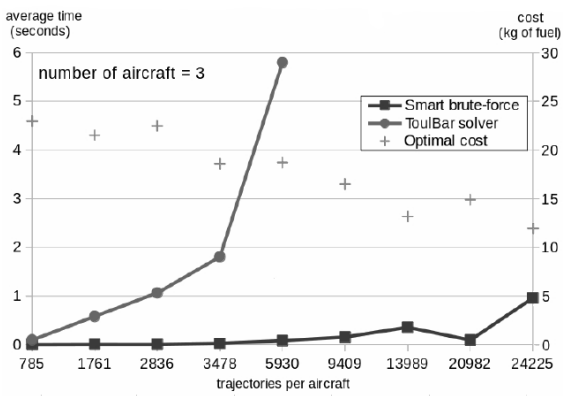


Figure 8: **Performance of the two solvers on trajectory generation without greedy algorithms.** Average solving time over 100 tests of the SBF (squares) and Toulbar2 (circles) approaches for instances of a Roundabout with 3 aircraft, function of the number of generated trajectories per aircraft. Crosses represent the average cost of the optimal solution.[5]

To overcome these limitations, the authors highlight a pruning method using a "greedy"

MinCostMaxClique[1] algorithm. Before launching the exact search, they use a fast, non-optimal greedy heuristic to find an initial valid solution. The cost calculated from this solution is then injected into Toulbar2 as an initial "upper bound". This trick allows the solver to prune the search tree much faster. It removes a considerable number of bad solutions and significantly accelerates the proof of optimality.

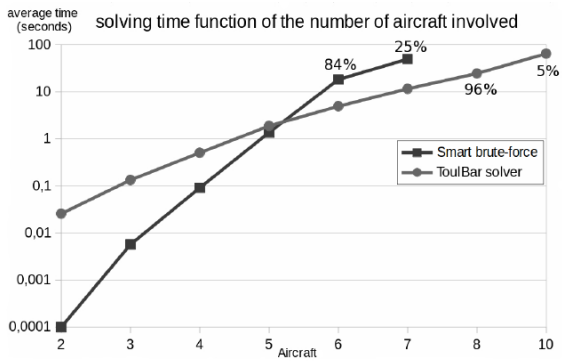


Figure 9: **Performance of the two solvers on trajectory generation with greedy algorithms.** Average solving time over 100 tests with statistical noise, of SBF (squares) and Toulbar2 (circles) approaches for instances of a Roundabout with approx. 850 trajectories per aircraft, function of the number of aircraft involved. When a method does not always succeed before the 60's time-out, its success rate is indicated.[5]

This use case shows that Toulbar2's performance depends heavily on the problem's topology. It also highlights the importance of hybridizing exact methods with fast, non-optimal heuristics. This combination guarantees response times that meet real-world industrial requirements.

3.2 Human resources and global constraints

Managing space and time is a major challenge. However, human resources planning introduces a different complexity. It requires modeling legal rules and individual preferences. This subsection highlights how WCSP research has had to evolve. It demonstrates the adaptation required to meet the expressivity needed to generate schedules.

Hospital staff scheduling is an optimization problem known for its high combinatorial complexity. In their study, Métivier et al. (2009)[10] apply the CFNs formalism and the Toulbar2 solver to generate optimal schedules for nurses. They must account for strict staff coverage requirements (hard constraints) as well as staff rest preferences (soft constraints).

The value of this document for our analysis lies in demonstrating the historical limitations of WCSPs for this type of problem. Historically, **CFN!**s (**CFN!**s) were designed around binary constraints. However, this type of constraint is poorly suited for modeling human rules. It would require creating an exponentially large constraint network. This makes the solution inefficient.

To overcome this issue, research around Toulbar2 focused on implementing soft global constraints. Unlike a binary constraint, a global constraint simultaneously evaluates a large sequence of assignments. For example, it can encompass a nurse's entire week or month in a single evaluation. Métivier et al. (2009) explain how specific algorithms, based on dynamic programming or flow networks, were integrated directly into Toulbar2. This integration allows the application of local consistencies (such as EDAC).

This article demonstrates a significant evolution. The WCSP paradigm is often associated with structural or biological problems. However, it has now achieved a much richer expressivity. Integrating soft global constraints into Toulbar2 proves its true capability. It can now successfully compete with traditional Constraint Programming (CP) solvers in the field of human scheduling.

3.3 Routing and networks

This final section addresses the allocation of purely abstract and logical resources. The networking and telecommunications field provides an excellent testing ground. It allows us to compare WCSPs against other major combinatorial optimization standards.

In their study, Lesaint et al. (2014)[8] examine a complex problem in telecommunications network allocation and configuration : service subscription conflict management. The issue is as follows. A customer may demand a combination of network services that are technically incompatible due to infrastructure limits. When this happens, the provider must find the best compromise. This involves downgrading the service or ignoring certain requested options. The goal is to minimize the overall penalty corresponding to customer dissatisfaction. This NP-hard decision problem was modeled as a WCSP.

The value of this article lies in its comparative analysis. Instead of isolating Toulbar2, the authors systematically compared the WCSP formulation against competing models. These included MILP and Partial Weighted Max-SAT. The results demonstrate the strength of the WCSP approach. By leveraging constraint propagation, it proves highly competitive for dynamically resolving these allocation conflicts.

This operations research use case provides empirical proof. WCSP modeling can compete with historically undisputed paradigms like MILP or Max-SAT for allocating abstract resources. Toulbar2 can sometimes saturate on certain purely linear problems. However, its modeling flexibility and computational power make it an effective alternative for resolving conflicting query satisfaction problems.

3.4 Conclusion

As seen in previous applications, the Toulbar2 solver excels at optimization and planning in deterministic environments. However, resource availability can become uncertain and dynamic. A prime example is IoT sensors, whose energy relies on random environmental conditions (Biason et al., 2017)[3]. In these situations, exact planning shows its limitations. It then becomes necessary to frame the problem using Markov Decision Processes. Here, Toulbar2 is no longer used as a standalone tool. Instead, it serves as an “internal computing engine” powering a probabilistic algorithm.
This need to hybridize exact constraint optimization with models that handle uncertainty marks a natural progression. It directly drives the integration of WCSPs into the broader fields of Artificial Intelligence and Machine Learning.

4 Artificial Intelligence & Machine Learning

We have seen that Toulbar2 is a leading exact solver for Cost Function Networks. We will examine how AI integrates with Toulbar2 by combining neural network perception with exact structural constraints. We will also explore research where machine learning improves upon or outperforms Toulbar2.

4.1 AI in the service of Toulbar2

There are instances where we have to face NP-hard combinatorial reasoning problems directly from natural inputs like images or unmapped protein structures. Traditional exact solvers cannot process raw sensory data and pure deep learning models struggle to perform exact logical reasoning. To resolve, a neural network processes the natural inputs and predicts pairwise cost matrices, the architecture presented on the paper trains using a dedicated loss function called the Emmental Pseudo-LogLikelihood, gaining time since this approach pushes the combinatorial solver out of the training loop entirely unlike other predecessors.
At inference, the neural network feeds the predicted Graphical Model directly into Toulbar2. It enforces the structural constraints and finds the exact optimal solution. The data flows through specific stages :

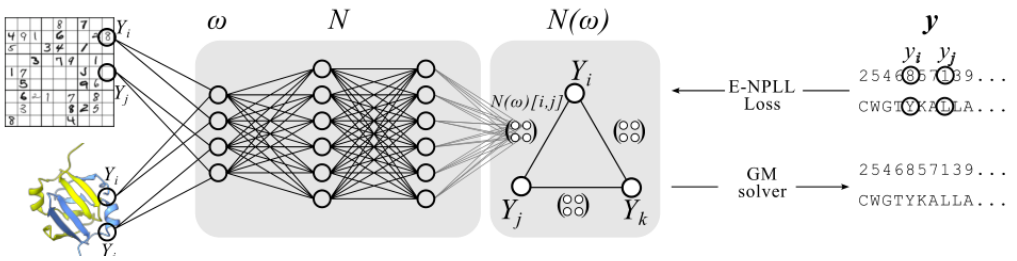


Figure 10: **Scaling Neuro-symbolic Problem Solving : Solver-Free Learning of Constraints and Objectives (Defresne et al., 2026)**.[6]

- Raw inputs enter the neural network.
- The network predicts all pairwise cost functions.
- Toulbar2 computes the exact discrete output.

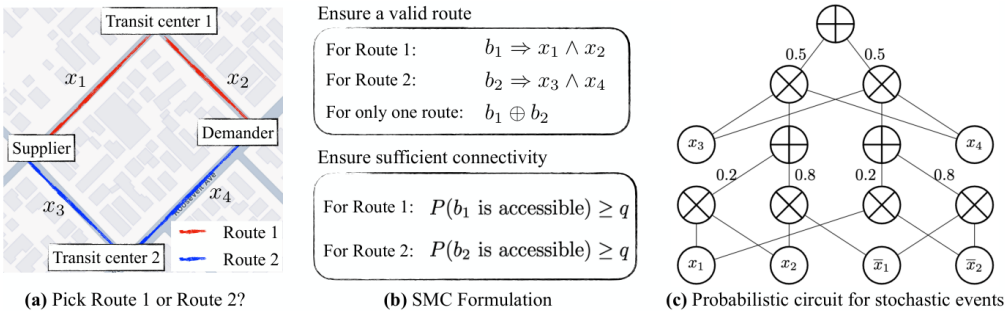
This effectively isolates perception from reasoning. It allows Toulbar2 to focus exclusively on the exact discrete optimization tasks where it dominates.

4.2 Yet New Ways Open

As established, Toulbar2 explores the whole solution space. This exploration makes it computationally infeasible for large scale constraint optimization problems. The application of modern artificial intelligence methods to directly address these specific limitations presents an interesting alternative.
Since Toulbar2 is too slow to generate optimal training labels for large instances, researchers have bypassed this limitation by utilizing self-supervised models like Deep Attentive Belief Propagation. This method infers optimal parameters dynamically without needing Toulbar2 to provide exact solutions.
We can also reduce communication overhead when integrating Toulbar2 with SAT solvers for *Satisfiability Modulo Counting* (SMC[2]). Traditional integrated systems run slowly due to continuous communication between the separate logic and probabilistic solvers, this problem

[2]SMC (Satisfiability Modulo Counting) : combines logic rules with probability. It's used in order to find answers that meet strict logical conditions and achieve a specific probability target.

is solved with integrated solvers like KOCO SMC[83], they improve this by tracking lower and upper bounds during probabilistic inference. They also evaluate partial variable assignments early in order to help to prevent exhaustive enumerations.



(a) Pick Route 1 or Route 2? **(b)** SMC Formulation **(c)** Probabilistic circuit for stochastic events

Figure 11: **Example of SMC : Robust Supply Chain with Different Network Size (Li et al., 2025)**.[9]

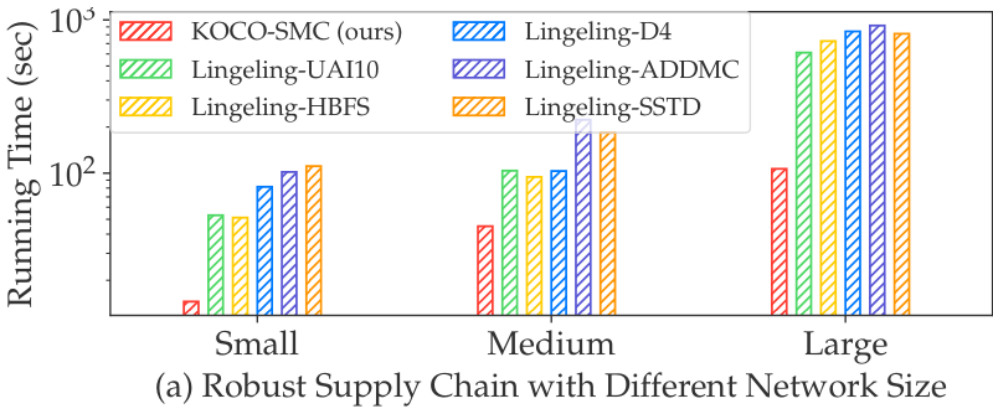


Figure 12: **Performance of each algorithm.**[9]

This algorithm is a significant improvement over previous systems like Lingeling-HBFS that rely on Toulbar2.
Artificial intelligence can additionally be utilized to match exact quality with linear scaling instead of exponential scaling Toulbar2 has. NeuroLifting addresses the limitations of Toulbar2 by using Graph Neural Networks to reparameterize decision variables. This transformation allows the optimization of models through standard gradient descent. This method matches the solution quality of Toulbar2 on moderate scales and outperforms it on large problems due to linear computational growth.

4.3 Conclusion

The integration of artificial intelligence resolves the scaling limitations of the Toulbar2 in some problems. Neural networks can also process the perception of the data to let the solver focus on discrete optimization tasks. Emerging algorithms bypass exhaustive search through the application of gradient descent and dynamic probabilistic tracking. This transition mandates that future solving methods rely on hybrid architectures rather than exact solvers.

[3]KOCO-SMC is a SMC solver by tracking lower and upper bounds in the probabilistic inference process.

5 Conclusion

6 Appendix

A Algorithms and Code Snippets

A.1 MinCostMaxClique algorithm

Algorithm 1: MinCostMaxClique

Where :

– p, the number of aircraft

– Vect, a *structure* representing a tuple of trajectories, and composed of an array *tab* of size p, containing the id of nodes from the p-partite graph ; an *index* ; and a *cost*

– H, a binary min heap of Vect sorted by their *cost*

```
Vect v; v';
v.tab[0..p − 1] := −1; v.index := 0;
v.tab[v.index] + +;
H.insert(v);
while H isn't empty do
    v := H.extract_top;
    if v is legal then
        if v is solution then
            return v;
        v' := v;
        v'.index + +; v'.tab[index] + +;
        H.insert(v');
        v.tab[index] + +;
    H.insert(v);
return "No solution";
```

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